

1 — Problem & Innovation

Commercial facilities can lose water through continuous leaks and abnormal usage that are difficult to prioritize from raw telemetry.

Innovation: context-aware anomaly detection combines occupancy, flow and usage behavior instead of relying only on a fixed leak threshold.

Outcome: Detect -> quantify wastage -> classify severity -> recommend action -> create maintenance ticket.

2 — Solution Architecture

IoT / Simulator -> Data Processing -> ML + Context Rules -> Incident Intelligence -> Dashboard + Ticket -> Facility Assistant

Telemetry: water flow, occupancy, flush count, sensor status. Detection: Isolation Forest + occupancy-aware occupancy-aware leak rule. Tech stack: Python, Streamlit, Pandas, NumPy, Scikit-learn.

3 — Working Prototype & AI

Reproducible scenarios include a continuous-flow leak-like event, abnormal high use and degraded sensor status.

Example: near-zero occupancy + about 3.8 L/min continuous flow -> anomaly -> critical severity -> estimated estimated wastage -> maintenance recommendation.

AI principle: deterministic detection plus AI/LLM-ready explanation and natural-language interaction.

4 — Impact & Deployment

Water: quantifies abnormal wastage for conservation. Operations: converts alerts into structured maintenance maintenance actions. Scalability: replace simulator with MQTT/API telemetry and connect to a CMMS. Future: Future: historical forecasting, adaptive baselines, fixture-level digital twins and live notifications.